203 Force and Motion (I)

2.3.1 · 2.3.2 Inertia and mass

Inertia is the tendency to maintain rest or constant velocity. Greater mass means greater inertia, so the same resultant force produces a smaller change in velocity per second. Inertia is a property, not an additional force. Without resistance, no continuing forward resultant force is needed to maintain constant velocity.

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2.3.3 Newton's first law

In an inertial frame, an object with zero resultant force remains at rest or moves with constant velocity in a straight line. Force changes velocity rather than maintaining it. An object moving forwards need not have a forward resultant force, and zero resultant force does not mean that every individual force is zero.

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2.3.4 · 2.3.5 Newton's second law

For constant mass, acceleration is determined by the resultant force and points in its direction. A greater resultant force gives greater acceleration for the same mass; a larger mass accelerates less under the same force. Draw a free-body diagram and combine all forces with signs before applying the law; do not substitute one force for the resultant.

$$a \propto F_{\text{net}} (m \text{ fixed})$$

$$a \propto \frac{1}{m} (F_{\text{net}} \text{ fixed})$$

$$\sum \vec{F} = m\vec{a}$$

a: acceleration magnitude (m s^{-2}); F_{net}: resultant force magnitude (N); m: mass (kg). Acceleration points along the resultant force.

Resultant force (N), not an individual force; m: constant mass (kg); a: acceleration (m s^{-2}). In an inertial reference frame.

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2.3.6 · 2.3.7 Mass and weight

Mass measures inertia, is expressed in kg and normally remains unchanged when location changes. Weight is gravitational force, measured in N, equal to mass times local gravitational field strength. On the Moon, mass is unchanged but weight is smaller. Weight has direction, and a scale's support-force reading need not equal weight.

$$\vec{W} = m\vec{g}$$

W: weight (N); m: mass (kg); g: gravitational field strength (N kg^{-1}). Weight points along the gravitational field.

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2.3.8 Terminal speed

As an object falls through a fluid, increasing speed raises the upward drag. Terminal speed is reached when drag plus upthrust balances weight, making resultant force and acceleration zero while the object continues falling. Only when upthrust is negligible can the balance be simplified to drag equals weight.

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2.3.9 Newton's third law

Two interacting objects exert simultaneous forces of equal magnitude and opposite direction on each other. The forces act on different objects, so they do not cancel in one object's free-body diagram. They cancel as internal forces only when both objects are included in a single system.

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